Chapter 2 – The Computer

The Computer

- Computer system consists of **input, output, memory, processing, networking, VR and paper interaction.**

Input Devices

- Used to **enter data into a computer.**
- Examples: Keyboard, mouse, touchscreen, scanner, sensors.
- Affect **efficiency** and **naturalness** of interaction.

Output Devices

- **Present information** to users.
- Examples: Monitor, projector, printer, speakers, VR headset.
- Affect **readability**, **usability** and **cognitive load**.

Virtual Reality (VR)

- **Simulates** an **immersive 3D environment.**
- Uses **headsets, motion controllers** and **tracking**.
- Important issues:
  - Immersion
  - Motion sickness
  - Spatial awareness

Physical Interaction

- Interaction **beyond keyboard and mouse.**

- Examples:
  - Voice input
  - Haptic feedback (vibration)
  - Biometric sensing

Paper Interaction

- Paper can be:
  - **Output (printing)**
  - **Input (scanning)**
- Connects physical and digital worlds.

Memory

- **RAM:** Temporary, fast, volatile.
- **Storage (SSD/HDD):** Permanent, slower.
- Low memory reduces system performance.

Processing

- Includes **CPU speed, network speed** and **response time.**
- **Faster response** generally **improves usability**.

Interactivity

1. **What goes in and out**
2. What the **system can do internally**

Batch Processing

- **Input given all at once.**
- No user control during execution.
- Errors require rerunning process.

Interactive Computing

- **Continuous interaction.**
- **Immediate feedback.**
- **User** remains in **control**.

Is Faster Always Better?

- No.
- **Too much speed can cause confusion and errors.**
- **Balance speed** with **understandability**.

Text Entry Devices

QWERTY Keyboard

- **Most common keyboard layout.**
- Designed to prevent typewriter jams.
- **Not optimized for speed.**

Dvorak Keyboard

- R**educes finger movement.**
- **Less fatigue**.
- **Can be faster than QWERTY.**
- **Common letters under strongest fingers**
- **Biased** towards **right hand!**

Chord Keyboard

- Characters entered using **key combinations.**
- **Small** and **portable**.
- **Fast after training.**

Phone Pad / T9

- **Multiple letters on one key.**
- T9 predicts intended words.
- **Requires repeated key presses**
- Surprisingly **fast with practice**
- Limited keys → **compact design**

Handwriting Recognition

- Uses **stylus** or **touchscreen**.
- Challenges:
  - Different writing styles
  - Joined letters
  - Messy handwriting

Speech Recognition

- Converts **speech into text.**
- Works best with:
  - Limited vocabulary
  - Trained users
- Problems:
  - Noise
  - Accents
  - Multiple speakers

Numeric Keypads

- Optimized for **entering numbers.**

- Used in **ATMs, calculators and phones.**

Pointing and Drawing Devices

Mouse

- Most common pointing device.
- **Indirect** device.
- **Fast, accurate** and easy to learn.
- Requires **hand-eye coordination.**

Mechanical Mouse

- Uses a **rotating ball.**
- Detects movement via **rollers**
- Works on most **flat surfaces**

Optical Mouse

- Uses **LED/laser.**
- More **accurate** and **reliable**.

Touchpad

- Common in laptops.
- **Detects finger movement**.

Trackball

- **Stationary device with rotating ball.**
- Saves desk space.

Joystick

- **Pressure** controls **cursor speed**.
- Used in games, aircraft and 3D navigation.

TrackPoint (**Keyboard Nipple**)

- **Mini joystick** on keyboard.
- Keeps hands on keyboard.

Touchscreen

- **Direct pointing device**.
- Fast and intuitive.
- Less precise for small targets.

Stylus

- **Pen-like** input device.
- Better precision than finger.

Light Pen

- Detects **screen light.**
- Rarely used today.

Digitizing Tablet

- Accurate drawing device.
- Used in graphics and CAD.

Eye Gaze

- **Cursor controlled by eye movement.**
- Useful for accessibility.
- **Expensive** for high accuracy
- Requires **headset** → if we want **high accuracy**

Cursor Keys

- **Arrow keys for movement.**
- Cheap but slow.

Discrete Positioning Controls

- **Fixed-step movement.**
- **not smooth continuous motion**
- Examples: D-pads, **remote controls.**

Direct Pointing

- **User touches target directly.**
- Example: Touchscreen.

Indirect Pointing

- **Separate device controls cursor.**
- Example: Mouse, joystick.

🎯 **SCENARIO-BASED EXAM GUIDE**

Scenario	Best Device	Reason
Photo editing with small icons	Mouse	Precision + drag support
Flight simulation	Joystick	Fluid, continuous movement
Tourist info kiosk	Touchscreen	Intuitive for walk-up users
Air traffic control	Light Pen/Stylus	Safety-critical precision pointing
VR training	Motion Controllers + HMD	3D immersion

Display Devices

Bitmap Displays

- **Images** made **from pixels.**

Resolution

- **Number of pixels on screen.**
- Higher resolution = more detail.

DPI (Dots Per Inch)

- **Pixel density.**
- Higher DPI = sharper images.

Aspect Ratio

- **Width : Height** ratio.
- Examples:
  - 4:3
  - 16:9
  - 21:9

Colour Depth

- Indicates **how many colours each pixel can display**.

Anti-Aliasing

- **Smooths jagged edges (**"jaggies") → **Diagonal lines look stair**-like because: Screens are made of **square pixels**
- **Anti aliasing** → **Softens edges by using shades of line colour**

CRT (Cathode Ray Tube)

- **Old display technology**.
- Heavy, bulky, can flicker.
- **Radiation concerns**.

Health Tips

- Avoid glare.
- Take breaks.
- Use proper lighting.

LCD

- **Lighter** and **smaller** than CRT.
- Less eye strain.
- Used in laptops, phones and TVs.

Special Displays

Vector Display / random scan

- **Draws lines directly.**
- Need constant redraw
- No jagged edges.

DVST - Direct View Storage Tube

- **No flicker.**

- Cannot erase selectively.

Large Displays

- Plasma screens.
- Video walls.
- Projectors.

Situated Displays

- Displays placed in **specific locations**.
- Meaning depends on **context** and **audience**.
- Examples:
  - Airport screens
  - Office door displays

Hermes

Small screens beside office doors.

- Visitors leave notes using stylus
- Office owner reads notes online
- Supports **asynchronous** communication.

Digital Paper

- Flexible electronic paper.
- Retains display without power.

Virtual Reality & 3D Interaction

3D Movement

6 Degrees of Freedom

- Translation: **X, Y, Z**
- Rotation: **Roll, Pitch, Yaw**

3D Input Devices

- 3D mouse
- Data gloves
- VR helmets
- Body tracking

VR Motion Sickness

Caused by **sensory conflicts**:

- **Head moves → image lags → brain conflict.**
- Brain detects **mismatch** → **nausea**

Types of VR

- Desktop VR
- Immersive VR
- CAVE systems

Dedicated Displays

Used when information **must always be visible and quickly readable**.

Analogue Displays

- **Dials** and **gauges**.
- Good for showing ranges.

Digital Displays

- **Numeric values**.
- Good for precision.

Head-Up Displays (HUD)

- Information shown in **user's field of view.**
- Common in aircraft.

Sound and Haptics

Sound

Sound is an **additional output channel**.

- Alerts
- Warnings
- Confirmation

Sound reduces need for visual attention

Haptic Feedback

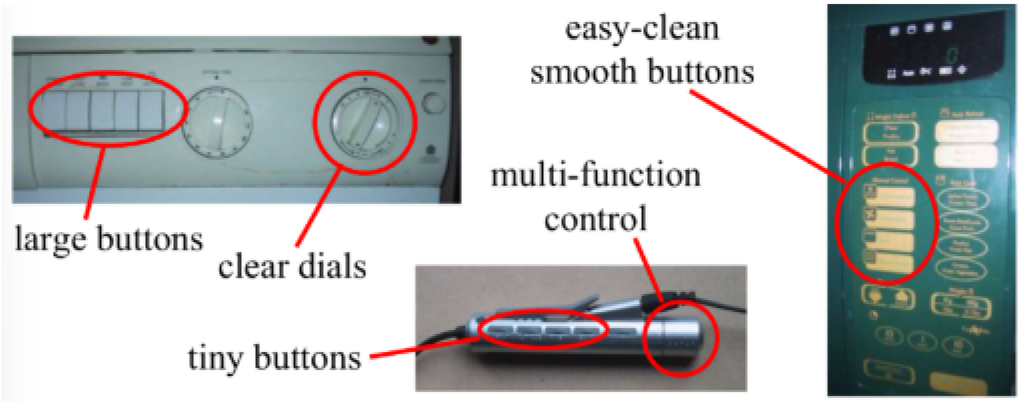
- **Vibration** or **force feedback**.
- Improves **realism** and interaction.

BMW iDrive

- Example of haptic feedback.
- Physical bumps confirm menu navigation.
- **small "bumps" felt while rotating**

Physical Controls

- Used in vehicles, appliances and industry.
- Should be clear and easy to use.



Environment & Bio-Sensing

Environmental Sensors

- Temperature
- Location
- RFID
- Weight sensors

Bio-Sensors

- Iris scanners
- Heart rate
- Body temperature
- Blink rate

Printing

- **Image** made from **small dots**

- Allows **printing** of **any character set** or **graphic**

Important Features

- Resolution (DPI)
- Speed (PPM)
- Cost → higher DPI - sharper o/p

Printer Types

- Dot-matrix
- Inkjet
- Laser
- Thermal

Fonts & Readability

Font Types

- Serif → better for print.
- Sans-serif → better for screens.

Pitch

- **Fixed pitch** → all characters **same width** (**Courier**)
- **Variable pitch** → different widths. all characters **same width** (**Courier**)

Readability

- Lowercase easier for reading.
- Uppercase better for short codes.

PDL & WYSIWYG

Page Description Language (PDL)

- Describes **page layout**.
- Example: PostScript.

WYSIWYG

- "What You See Is What You Get."
- **Screen and printed output** are **similar** but not identical.

Scanners & OCR

Scanner

- Converts **paper** into **bitmap images.**

OCR - Optical Character Recognition

- Converts **bitmap images** into **editable text.**
- Challenges:
  - Fonts
  - Noise
  - Complex layouts

Paper-Based Interaction

- Paper can act as **input** through **scanning and OCR**

Storage

RAM

- **Fast.**
- **Temporary.**
- **Volatile**.

Magnetic Disks

- **Hard disks**, floppy disks.
- **Permanent storage**.

Optical Disks

- **CDs** and **DVDs**.

Flash Memory

- **Non-volatile**
- Used in USB drives and phones.

Virtual Memory

- Uses **disk space as extra RAM.**
- Makes system slower.

Compression

Lossless

- No data lost.
- Examples: ZIP, GIF.

Lossy

- Some data removed.
- Examples: JPEG, MP3.
- **removes details humans barely notice**

Storage Formats

Text

- ASCII
- UTF-8
- RTF
- SGML
- XML

Media

- JPEG, GIF, TIFF
- MPEG, WAV, MOV

Information Access

Indexing

- **Speeds up searching.**

Methods

- **Exact match**
- **Soundex**
- **Free-text indexing**

    - index every word
    - flexible searching
    - requires lots of storage

Processing & Performance

Finite Processing Speed

- Systems have **limits**.
- Design must **match human speed.**

Moore's Law

- Processing power roughly doubles every 18 months.

Myth of Infinitely Fast Machine

- **Delays** always **exist**.
- Interfaces **must provide feedback.**

Performance Bottlenecks

- CPU bound
- Storage bound
- Graphics bound
- Network bound

Networked Computing

Provides:

- **Shared resources**
- **Communication**

- **Shared processing**

Issues

- **Delays**
- Conflicts
- Unpredictability

Internet

History

- Started as ARPANET (1969).

Core Protocols

- TCP → Data transfer.
- IP → Addressing and routing.

Higher-Level Services

- HTTP
- SMTP (Email)
- Web services

Layered Architecture

- Simple protocols at the bottom.
- Complex services built on top.